

Exercise #6

Introduction To Artificial Intelligence

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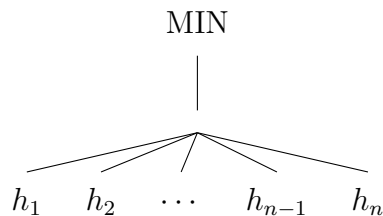
Due Date: March 13th, 2018

1 Suboptimal Min vs Optimal Min

Problem Statement. *Prove the following assertion: For every game tree, the utility obtained by MAX using minimax decisions against a suboptimal MIN will be never be lower than the utility obtained playing against an optimal MIN. Can you come up with a game tree in which MAX can do still better using a suboptimal strategy against a suboptimal MIN?*

We will do so by induction.

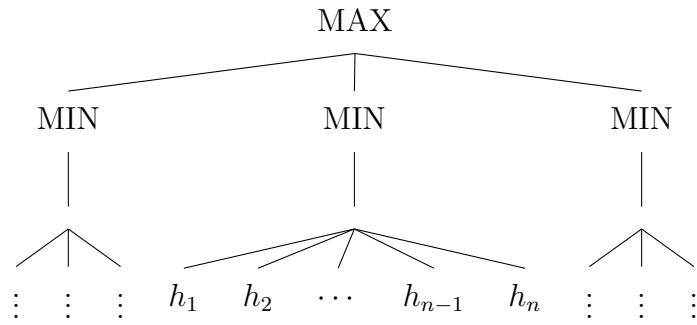
Basis Case The base case is a *MIN* node with all terminals. Suppose we have the following leaf such that for any node h_x , they are ordered in ascending order based on the utility value $h_1 \leq h_2 \leq \dots \leq h_{n-1} \leq h_n$:



An optimal *MIN* would always pick h_1 , while a suboptimal would pick from $\{h_1, h_2, \dots, h_{n-1}, h_n\}$. From this, we can conclude

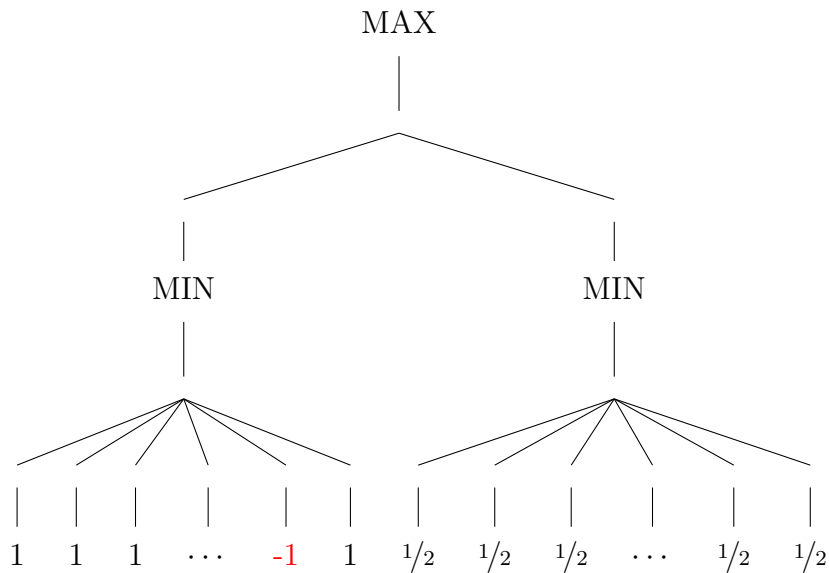
$$\text{Optimal } MIN\text{'s Utility} \leq \text{Suboptimal } MIN \text{ Utility} \quad (1)$$

Inductive Case The recursive case is for *MIN* with parents and children.



However, because the game tree can be reduced to a *MIN* node with all terminals (such as the base case), our inductive step holds.

For a game tree of a suboptimal *MAX* vs a *MIN*, we can produce the following chess game:



An optimal *MAX* would pick the rightmost *MIN*, which would result in only a draw; however, a suboptimal *MIN* would only pick the leftmost *MIN*, resulting in in almost all wins, with only one possibility of losing.